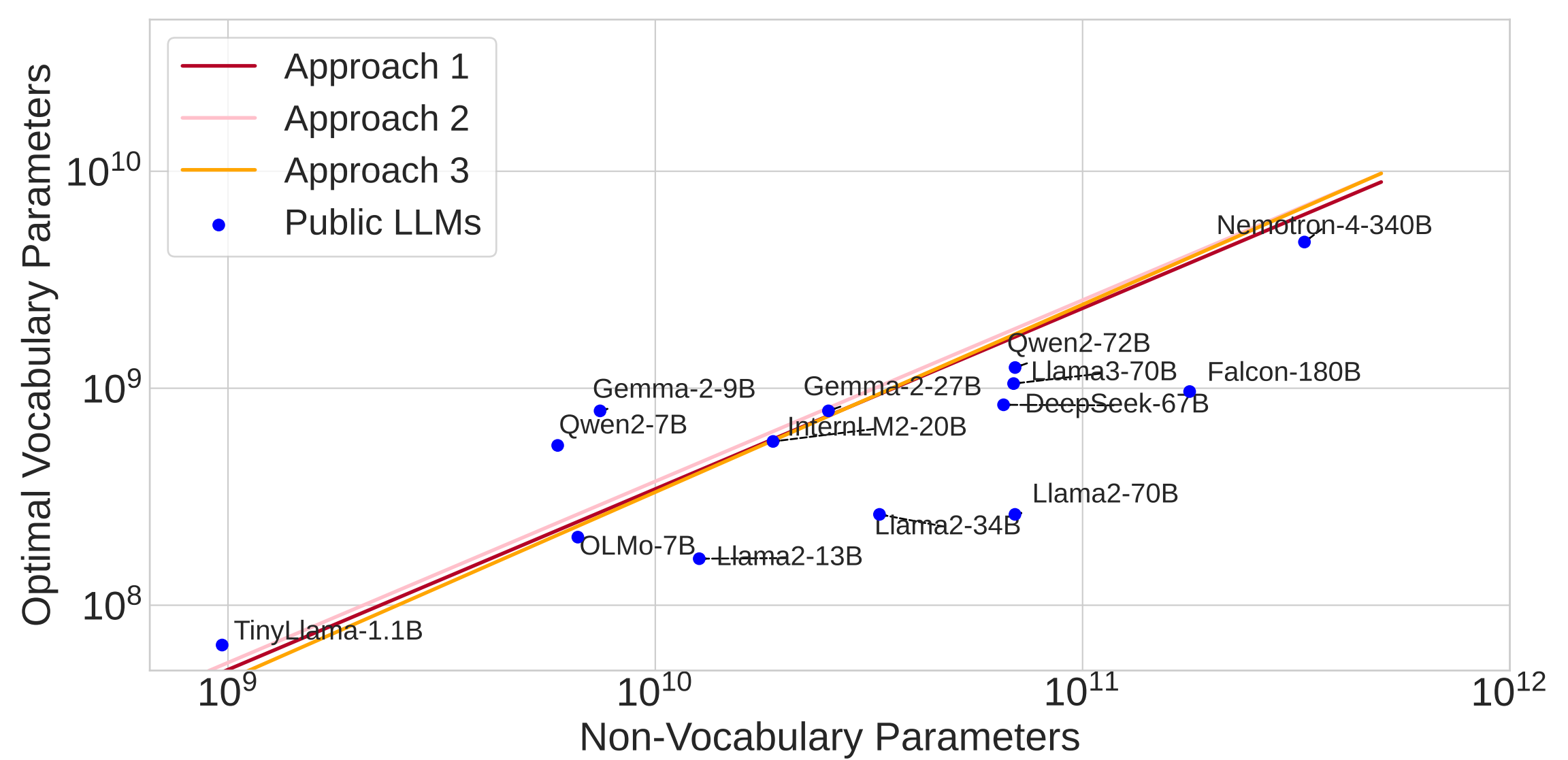
Approach 1
Approach 2
Approach 3
Public LLMs
Optimal Vocabulary Parameters
Non-Vocabulary Parameters
10^{8}
10^{9}
10^{10}
10^{9}
10^{10}
10^{11}
10^{12}
TinyLlama-1.1B
Gemma-2-9B
Qwen2-7B
OLMo-7B
Llama2-13B
Gemma-2-27B
InternLM2-20B
Llama2-34B
Llama2-70B
Qwen2-72B
Llama3-70B
DeepSeek-67B
Falcon-180B
Nemotron-4-340B